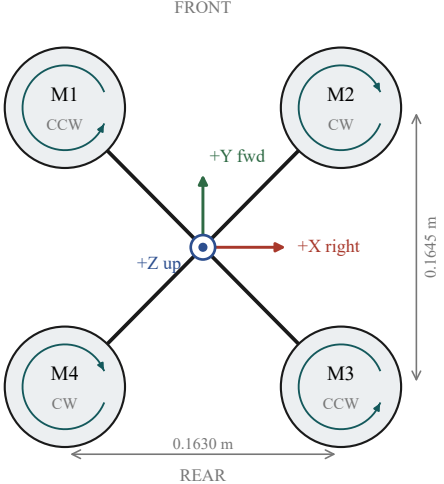
(a)



(b)

axis	M1	M2	M3	M4	produced by
Thrust T	+	+	+	+	$k_F \Sigma u_i$
Pitch τ_x	+	+	−	−	$k_F a_y (u_1+u_2-u_3-u_4)$
Roll τ_y	+	−	−	+	$k_F a_x (u_1-u_2-u_3+u_4)$
Yaw τ_z	−	+	−	+	$-k_M (u_1-u_2+u_3-u_4)$

Filled = that rotor speeds up. $u_i = \omega_i^2$; $a_x = 0.0815$ m, $a_y = 0.08225$ m. A CCW rotor is driven against aerodynamic drag, so the airframe receives the opposite reaction. Yaw is therefore the only row set by k_M instead of k_F, and $k_M/k_F \approx 0.01$ m -- an eightfold authority deficit.

Sensor lever arms on the $+Y$ centreline: PMW3901 at $+0.040$ m, VL53L0X at $+0.070$ m from the airframe centre.